



MBA (Trimester-II) Regular Examination February 2026

Roll no.....

Name of the Course and semester: MBA, Semester II

Name of the Paper: Business Analytics

Paper Code: MBA 206

Time: 3 hours

Maximum Marks: 100

Note:

- (i) This question paper contains two sections: Section A and Section B
- (ii) Both Sections are compulsory
- (iii) Answer any two sub-questions from a, b & c in each main question of Section A. Each sub-question carries 10 marks.
- (iv) Section B, consisting of a case study, is compulsory. It is 40 Marks.

Section A

Q1. (2X10=20 Marks)

a. A company believes that the average delivery time of its orders follows a normal distribution with a mean of 5 days. Management wants to test whether recent process changes have reduced the average delivery time. Explain how you would formulate the null and alternative hypotheses for this situation and justify the use of the normal distribution in this context.

CO1, CO2

b. Explain the role of descriptive analytics in managerial decision-making. How do summary measures such as totals, averages, percentages, and trends help managers understand business performance and identify areas requiring attention?

CO1, CO2

c. A manufacturer produces soft drinks and sell it in cans of 300 ml. To test if the filling machine is dispensing 300 ml of fluid, a test was conducted. A sample of 16 cans was collected, and the average value of fluid filled in each can for the sample was 299.6 ml, with a sample standard deviation of 0.585 ml. Test the hypothesis at an α of 0.05. Refer to Annexure III if required.

(CO2, CO3, CO4)

Q2. (2X10=20 Marks)

a. A company tests whether a new marketing campaign increases average monthly sales. Explain what a Type I error and a Type II error would represent in this context. Which error would be more costly for management, and how might this influence the choice of significance level (α)?

(CO2, CO3)

b. Why is a two-population t-test (either pooled or unpooled) not appropriate for testing differences among three or more populations? In your answer, explain the statistical reasoning related to hypothesis structure, error rates, and interpretation of results, and justify which test is preferred in such situations.

(CO2, CO3, CO4)



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C.

Gulfstream Aerospace Company produced three different prototypes as candidates for mass production as the company's newest large-cabin business jet, the *Gulfstream IV*. Each of the three prototypes has slightly different features, which may bring about differences in performance. Therefore, as part of the decision-making process concerning which model to produce, company engineers are interested in determining whether the three proposed models have about the same average flight range. Each of the models is assigned a random choice of 10 flight routes and departure times, and the flight range on a full standard fuel tank is measured (the planes carry additional fuel on the test flights, to allow them to land safely at certain destination points). Range data for the three prototypes, in nautical miles (measured to the nearest 10 miles), are as follows.²

Prototype A	Prototype B	Prototype C
4,420	4,230	4,110
4,540	4,220	4,090
4,380	4,100	4,070
4,550	4,300	4,160
4,210	4,420	4,230
4,330	4,110	4,120
4,400	4,230	4,000
4,340	4,280	4,200
4,390	4,090	4,150
4,510	4,320	4,220

Do all three prototypes have the same average range? Construct an ANOVA table, and carry out the test. Explain your results.

Refer to Annexure IV if required.

(CO2, CO3, CO4)



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Q3.

(2X10=20 Marks)

- a. A telecom firm states that the average call drop rate per day is **less than or equal to 2%**. A sample of **100 days** shows a mean call drop rate of **2.3%**. The population standard deviation is **1%**. Test the firm's claim at a **1% level of significance**.

Note. Critical Z-value = **2.33**

CO1, CO2, CO3

- b. Compare the Moving Average method with Single, Double, and Triple Exponential Smoothing as forecasting techniques. From a managerial perspective, explain when each method would be most appropriate and how the presence of trend and seasonality influences the choice of forecasting technique.

CO5

- c. In a campaign against a disease, a medicine was administered to 500 people out of a total population of 2000. The number of people with symptoms of the disease after the experiment is as illustrated below. Discuss the usefulness of the medicine at 95% CI.

Treatment	Symptoms persist	Symptoms do not persist	Total
Medicine administered	20	480	500
Medicine not administered	100	1400	1500
Total	120	1880	2000

Note: χ^2 (table @ DF 1 and $\alpha=0.05$) = 3.841

(CO3, CO4)



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Section B

Q4. Case Study

(40 Marks)

You are required to design a mini research project that will help you understand and apply the concepts of hypothesis testing and statistical tools covered in the course (such as z-test, t-test, chi-square test, ANOVA, correlation, etc.). The project should be practical in nature and based on a realistic business, economic, social, or policy-related problem. The project design must clearly demonstrate how hypothesis testing would be applied. Your project proposal should be structured and discussed under the following headings:

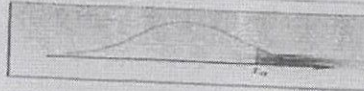
- Title of the Project
- Context of the Study (briefly describing the background of the problem)
- Broader Research Questions and Key Objectives of the Project. Clearly state one main research question, two to three supporting research questions
- Conceptual Model to be Analyzed, explaining the key independent variable(s) and dependent variable(s), the expected relationships among variables (diagrammatic representation is encouraged)
- Data Collection Plan (**Target Population:** Who are the respondents or units of analysis?, **Sampling Approach:** How will the sample be selected? **Data Collection Strategy,** Type of measurement scales used (nominal, ordinal, interval, ratio))
- Analysis of Research Questions. For each research question, clearly specify:
 - The null hypothesis (H_0) and alternative hypothesis (H_1)
 - The appropriate statistical test to be used
 - Justification for selecting the test
 - The expected managerial or decision-making implication of the result

CO1, CO2, CO 3, CO4, CO5, CO6



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Annexure III

TABLE 3 Critical Values of the t-Distribution



Degrees of Freedom	$t_{.100}$	$t_{.050}$	$t_{.025}$	$t_{.010}$	$t_{.005}$
2	1.638	1.960	2.306	2.920	3.182
3	1.638	1.960	2.306	2.920	3.182
4	1.533	1.833	2.179	2.776	3.041
5	1.476	1.753	2.015	2.571	2.876
6	1.440	1.699	1.943	2.447	2.800
7	1.415	1.650	1.895	2.365	2.747
8	1.397	1.601	1.850	2.306	2.689
9	1.385	1.553	1.812	2.262	2.650
10	1.372	1.512	1.771	2.228	2.617
11	1.363	1.482	1.746	2.201	2.591
12	1.356	1.460	1.725	2.179	2.576
13	1.350	1.445	1.708	2.160	2.561
14	1.345	1.433	1.694	2.145	2.548
15	1.341	1.423	1.682	2.131	2.536
16	1.338	1.415	1.671	2.119	2.525
17	1.335	1.408	1.661	2.109	2.516
18	1.333	1.402	1.652	2.100	2.508
19	1.331	1.397	1.644	2.092	2.501
20	1.329	1.393	1.637	2.085	2.495
21	1.328	1.389	1.631	2.079	2.489
22	1.326	1.386	1.626	2.074	2.484
23	1.325	1.383	1.621	2.069	2.479
24	1.324	1.381	1.617	2.065	2.475
25	1.323	1.379	1.614	2.061	2.471
26	1.322	1.377	1.611	2.058	2.468
27	1.321	1.375	1.608	2.055	2.465
28	1.320	1.373	1.605	2.052	2.462
29	1.319	1.371	1.603	2.050	2.460
30	1.318	1.369	1.601	2.048	2.458
40	1.314	1.364	1.596	2.043	2.452
60	1.310	1.359	1.591	2.038	2.447
120	1.306	1.355	1.587	2.034	2.443
∞	1.303	1.351	1.583	2.030	2.439

Source: R. A. Mendenhall, "Tables of Percentage Points of the t-Distribution," Biometrika 32 (1944), p. 198.
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Annexure IV
Anova: Single Factor

SUMMARY

Groups	Count	Sum	Average	Variance
Type A	10	44070	4407	11023
Type B	10	42410	4241	12366
Type C	10	41350	4135	5183

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	375920	2	187960	19.74	0.00	3.35
Within Groups	257150	27	9524			
Total	633070	29				



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Table. Chi-Square Distribution

TABLE 4 (continued) Critical Values of the Chi-Square Distribution

Degrees of Freedom	1% 0.01	5% 0.05	10% 0.10	20% 0.20	50% 0.50	90% 0.90
1	2.70554	3.84146	5.02389	6.63489	6.63489	7.87944
2	4.60517	5.99147	7.37776	9.21034	9.21034	10.5966
3	6.25139	7.87815	9.34840	11.3449	11.3449	12.8381
4	7.77944	9.48773	11.1454	13.2767	13.2767	14.8602
5	9.23635	11.0705	12.8325	15.0863	15.0863	16.7506
6	10.5966	12.5916	14.4494	16.8119	16.8119	18.5476
7	12.0170	14.0671	16.0128	18.4753	18.4753	20.2777
8	13.3616	15.5073	17.5346	20.0902	20.0902	21.9550
9	14.6837	16.9190	19.0228	21.6660	21.6660	23.5893
10	15.9872	18.3070	20.4831	23.2093	23.2093	25.1882
11	17.2750	19.6751	21.9200	24.7250	24.7250	26.7569
12	18.5494	21.0261	23.3367	26.2170	26.2170	28.3062
13	19.8119	22.3621	24.7356	27.6883	27.6883	29.8194
14	21.0643	23.6848	26.1190	29.1411	29.1411	31.3193
15	22.3027	24.9958	27.4883	30.5779	30.5779	32.8013
16	23.5418	26.2962	28.8450	31.9994	31.9994	34.2672
17	24.7690	27.5871	30.1910	33.4087	33.4087	35.7183
18	25.9894	28.8693	31.5269	34.8053	34.8053	37.1564
19	27.2036	30.1435	32.8523	36.1908	36.1908	38.5822
20	28.4120	31.4104	34.1696	37.5662	37.5662	39.9968
21	29.6151	32.6705	35.4789	38.9321	38.9321	41.4010
22	30.8133	33.9244	36.7807	40.2894	40.2894	42.7956
23	32.0069	35.1725	38.0757	41.6384	41.6384	44.1813
24	33.1955	36.4151	39.3641	42.9798	42.9798	45.5585
25	34.3816	37.6525	40.6455	44.3141	44.3141	46.9278
26	35.5631	38.8852	41.9232	45.6412	45.6412	48.2899
27	36.7412	40.1133	43.1974	46.9630	46.9630	49.6449
28	37.9159	41.3372	44.4667	48.2782	48.2782	50.9933
29	39.0875	42.5569	45.7322	49.5870	49.5870	52.3356
30	40.2560	43.7729	46.9929	50.8922	50.8922	53.6720
40	51.8050	55.7585	59.3417	63.6907	63.6907	66.7659
50	63.1671	67.5048	71.4202	76.1539	76.1539	79.4890
60	74.3970	79.0819	83.2976	88.3794	88.3794	91.9517
70	85.5271	90.5312	95.0231	100.422	100.422	104.215
80	96.5782	101.879	106.629	112.329	112.329	116.321
90	107.565	113.115	118.136	124.116	124.116	128.299
100	118.498	124.342	129.561	135.807	135.807	140.169

Source: C. M. Thompson, "TABLES of the Percentages Points of the χ^2 -Distribution," *Biometrika*, 32 (1945), pp. 145-59. Reproduced by permission of the Cambridge Service.